

Decision Model Selection and Cross-Validation with an MLP

Comparing k-fold, leave-one-out, grid search, randomised search, and halving search to select the best MLP on the Iris dataset.

Jibril Hussaini | Python · scikit-learn · NumPy · matplotlib · pandas

Overview

Choosing the right model configuration is as important as the model itself. This project investigates cross-validation strategies and hyperparameter search algorithms for a multi-layer perceptron (MLP) classifier trained on the Iris dataset (150 samples, 4 features, 3 classes). It works through eleven focused questions that build from basic k-fold CV up to advanced search strategies including randomised and successive-halving grid search.

Goals

Demonstrate how cross-validation accurately estimates generalisation error and how automated hyperparameter search identifies the best MLP architecture and regularisation strength without touching the held-out test set.

Objectives

- Visualise pairwise feature scatter plots for train and test splits to understand class separability.
- Measure how k-fold CV mean and variance change as k grows from small to leave-one-out.
- Overlay test-set accuracy on the CV curve to verify that cross-validation tracks true generalisation error.
- Run GridSearchCV over hidden_layer_sizes and alpha with a 3-D score surface and quantify the number of models trained.
- Compare RandomizedSearchCV (log-uniform alpha) and HalvingGridSearchCV against exhaustive grid search for speed and final accuracy.

Approach

The Iris dataset is split into a training set (75 samples) and a held-out test set (75 samples). Each cross-validation experiment is conducted exclusively on the training set; the test set is only evaluated once at the end. For grid search, the parameter grid spans 6 hidden-layer sizes and 4 alpha values (24 combinations, each validated with 5-fold CV, yielding 120 fits plus one refit). Randomised search samples log-uniformly over alpha and discretely over layer sizes. Halving search prunes weak candidates early by doubling the data budget at each successive round.

Outcomes

- K-fold CV mean accuracy stabilises quickly with growing k, while variance increases markedly at large k due to tiny per-fold validation sets.
- Leave-one-out CV produced a comparable mean to k-fold but with the highest per-fold variance

and the greatest computational cost.

- GridSearchCV identified the best MLP configuration; the score surface was flat across most of the grid, confirming low sensitivity to these hyperparameters on Iris.
- RandomizedSearchCV matched the full grid score while evaluating far fewer combinations, and HalvingGridSearchCV reached the same final accuracy faster by discarding weak candidates early.